

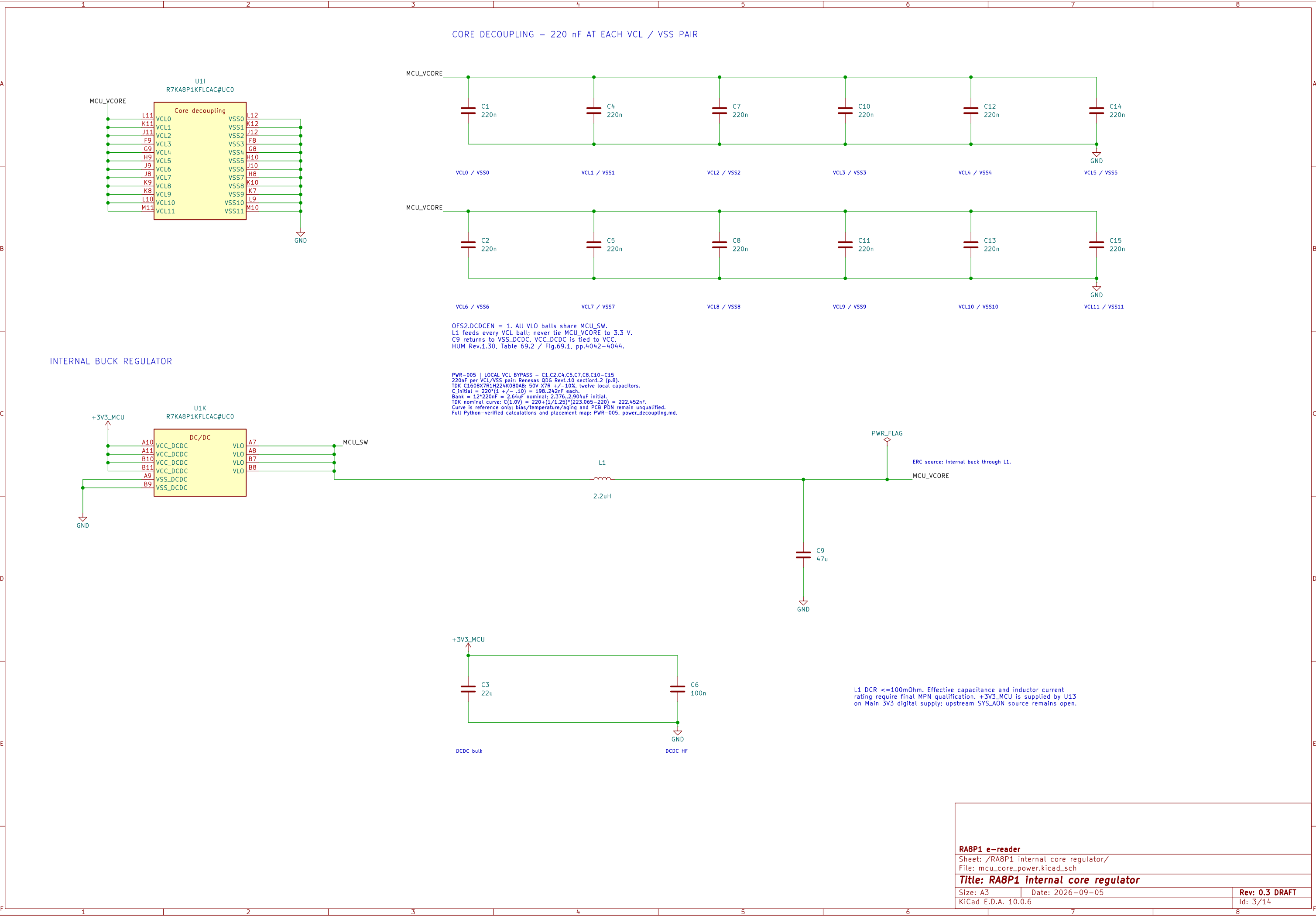
DRAFT: Existing processor units retained. Signal allocation and clocks/debug wiring are not yet complete.

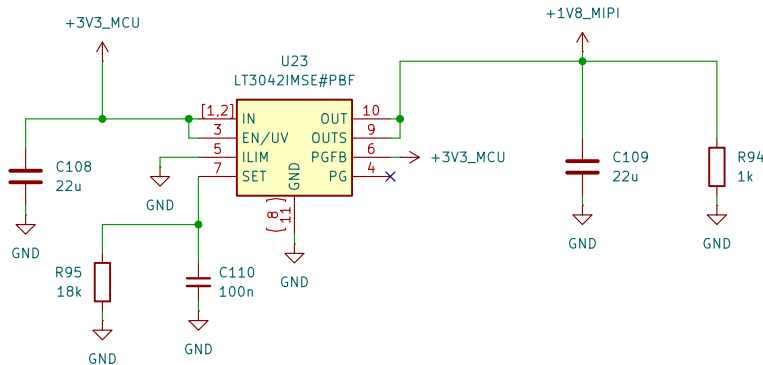
The diagram illustrates the IO allocation for the RABP1 e-reader. It includes several key components and their connections:

- Power and Grounding:** +3V3_MCU and +3V3_USBHS_A are connected to various components. A 120R resistor (FB1) is used for the +3V3_MCU supply. A 10uF capacitor (C63) is connected to the +3V3_USBHS_A supply.
- USB Interface:** The USB-004 module is connected to the +3V3_MCU and +3V3_USBHS_A supplies. It includes a 120 ohm resistor (FB1) and a 100 MHz capacitor (C63). The module is connected to the +3V3_MCU and +3V3_USBHS_A supplies.
- SDRAM:** The SDRAM module is connected to the +3V3_MCU and +3V3_USBHS_A supplies. It includes a 120 ohm resistor (FB1) and a 100 MHz capacitor (C63). The module is connected to the +3V3_MCU and +3V3_USBHS_A supplies.
- GPIOs:** The GPIO module is connected to the +3V3_MCU and +3V3_USBHS_A supplies. It includes a 120 ohm resistor (FB1) and a 100 MHz capacitor (C63). The module is connected to the +3V3_MCU and +3V3_USBHS_A supplies.
- Other Components:** The diagram also shows connections for the +3V3_MCU, +3V3_USBHS_A, and various other components like the +3V3_MCU, +3V3_USBHS_A, and various other components.

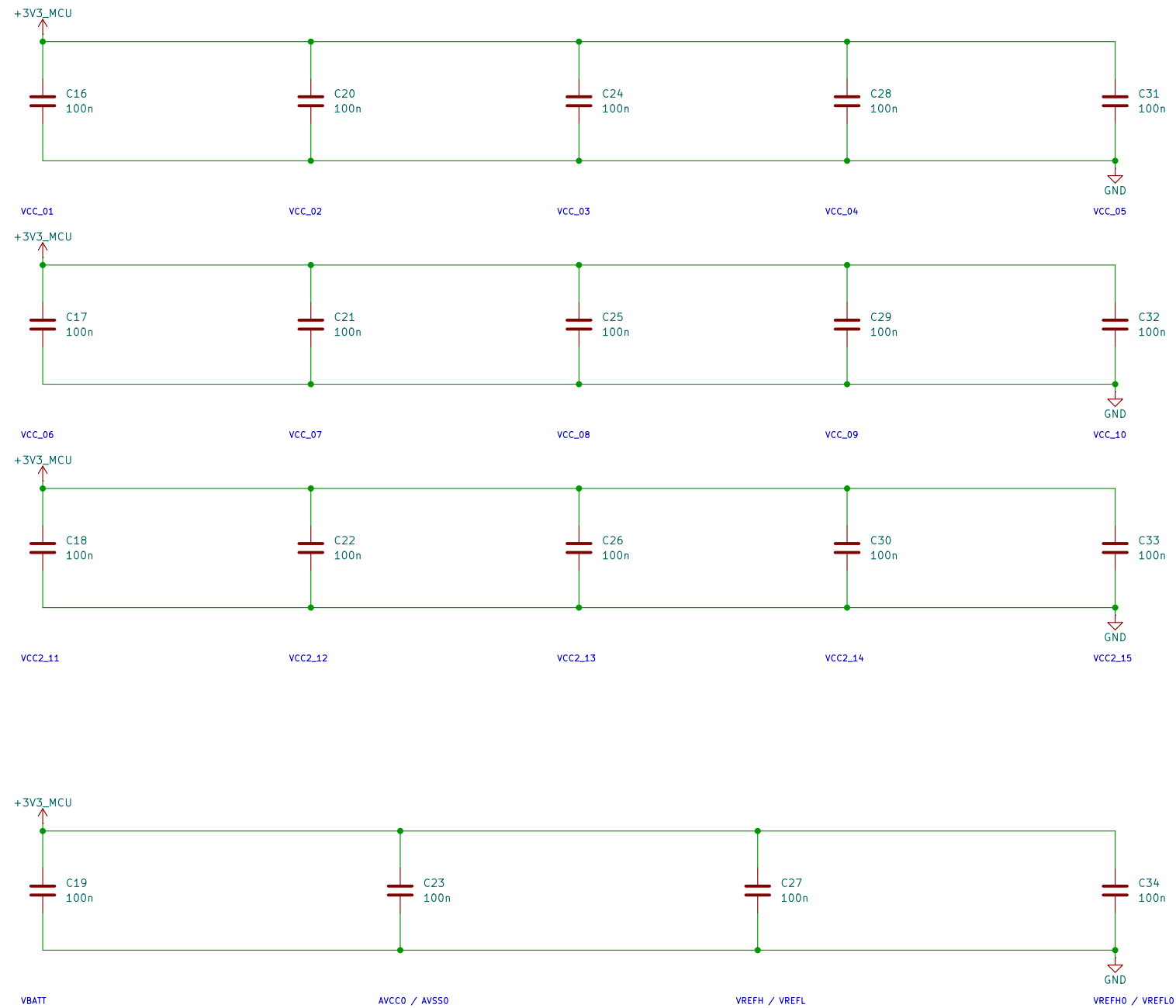
The diagram is a detailed schematic showing the electrical connections between the RABP1 e-reader and its various components. It includes a title block with the following information:

- Title:** RABP1 IO allocation
- Sheet:** /RABP1 IO allocation/
- File:** mcu_interfaces.kicad_sch
- Size:** A2
- Date:** 2026-09-05
- Rev:** 0.3 DRAFT
- Id:** 2/14



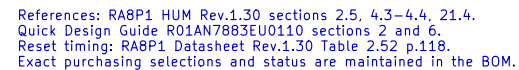
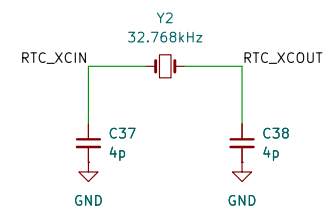
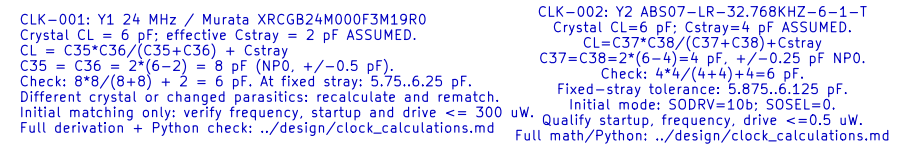


I/O DECOUPLING – 100 nF AT EACH NUMBERED VCC / VSS PAIR

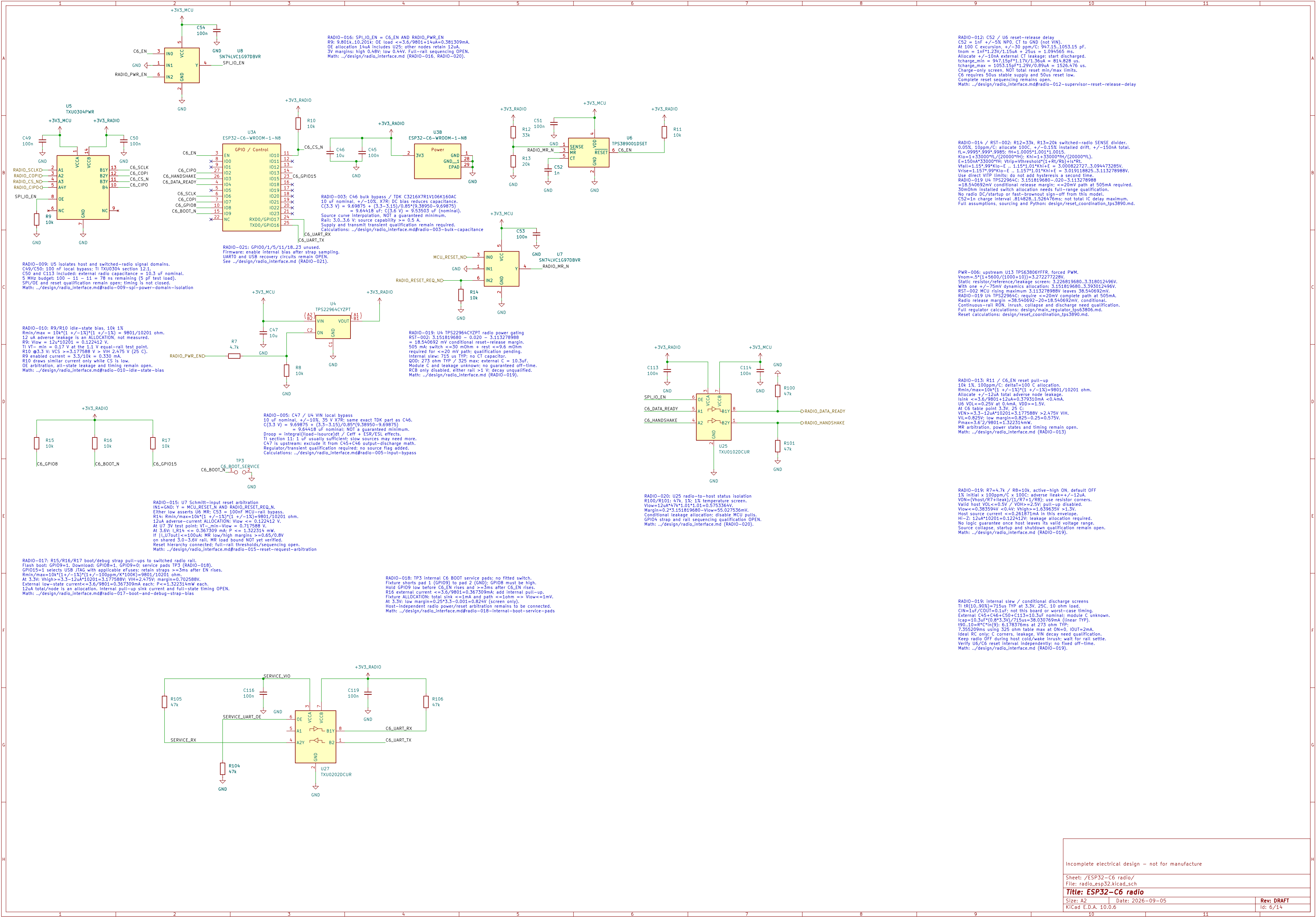


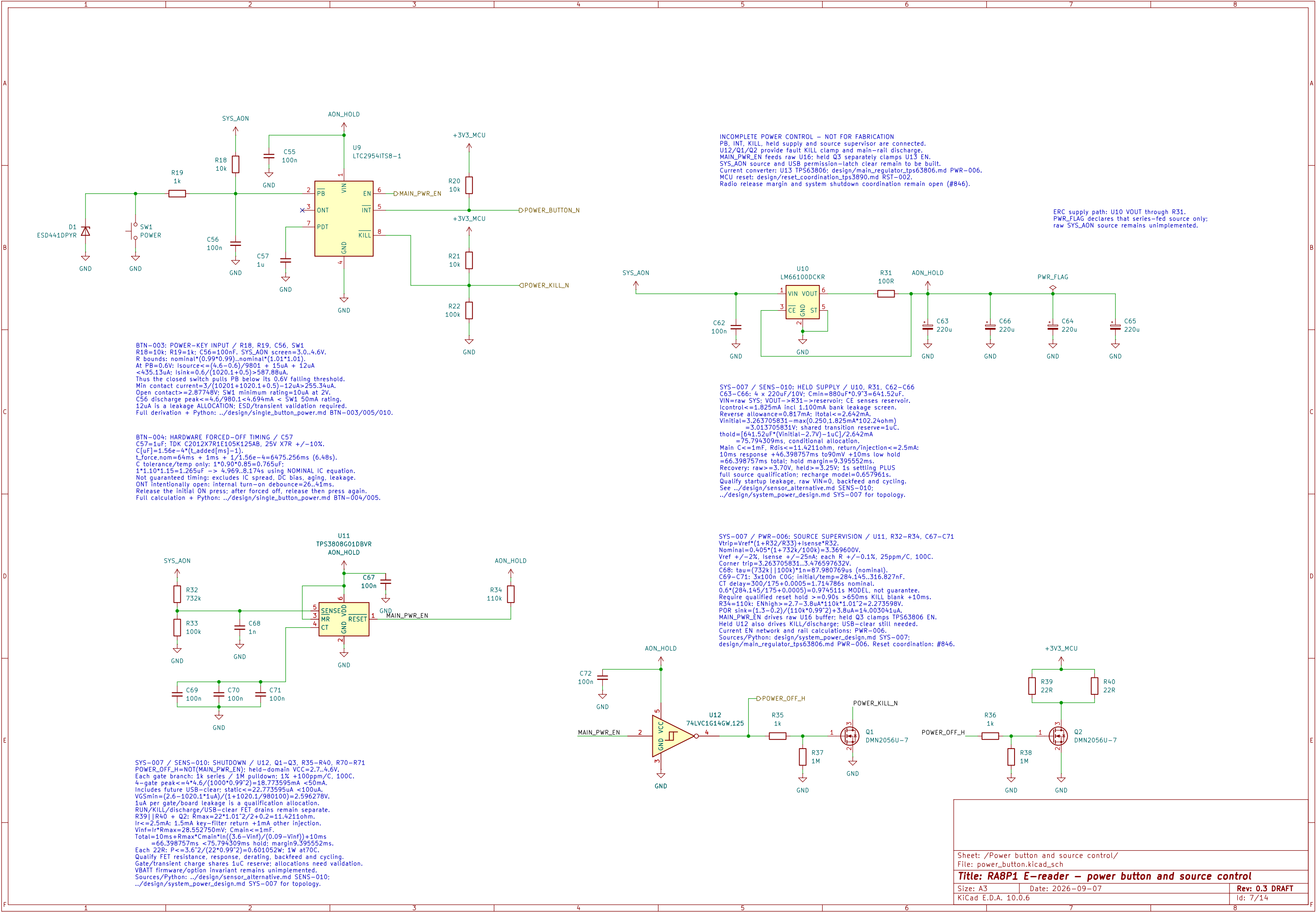
CMS-013; +20mA steady CSI branch planning allocation
 4.1mA PHY + 1.989593mA bleed + 13mA allocated GND
 + 0.102mA SET + 0.100mA auxiliary = 19.291593mA.
 13mA is an engineering envelope, not an interpolated guarantee.
 Auxiliary allowance requires verification; startup/inrush excluded.
 AVCC/MPI remains within existing MCU allocation.
 Main subtotals: 2.075 + 0.020 = 2.095A; full system budget pending.
 At 24VDC input, 3.2V input, assuming 75% efficiency:
 Input 2.961817158A; converter loss 2.369453727W.
 Source, current and thermal acceptance remain open.
 Sources, conditions and Python: design/camera_storage_interfaces.md.

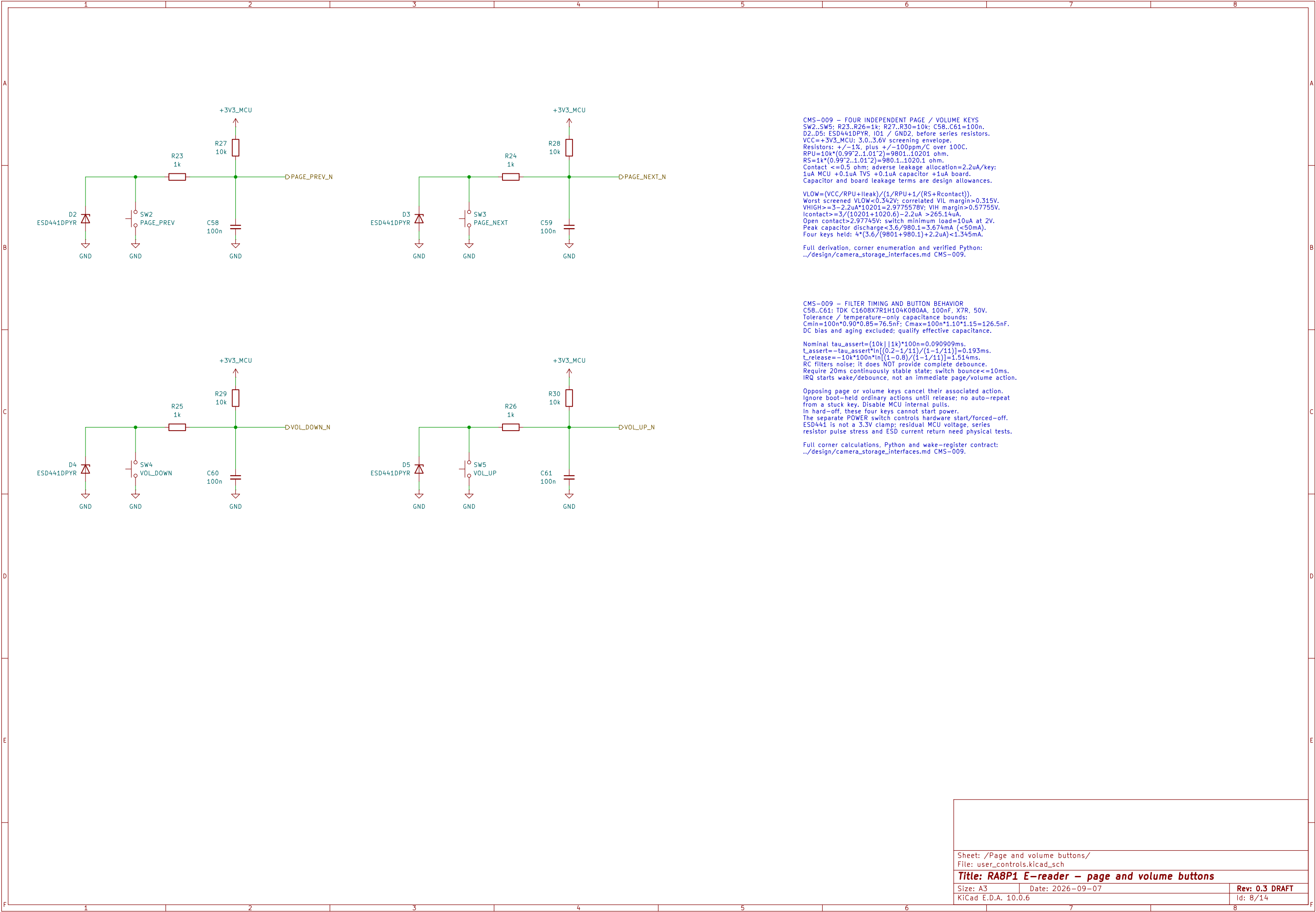
3



```
RST-002: U2 TPS3890U1DSET; R67=33k, R74=20k, R75=10k, C95=10n.  
Sense resistors: +0.05%, OhmCo over 100C, allocated + / -0.15% assembly/aging.  
Klo=9995.99*Klo-E = 9.9902E+25; E=1.0005E+10; Klo=330000.00*Fh; Fh=330000.00*Fh;  
Klo=4.133000e/f/(200000*f.h); E=1.33000e/f/(20000*f.h); E=150nA/330000f.h.  
Vfall=1.15*.99*Klo-E = 1.15*1.01*Khi+E = 3.00082727Z.3.094473285V.  
Vrise=1.157*.99*Klo-E = 1.157*1.01*Khi+E = 3.01911882S.3.113278988V.  
Main release margin=3.151819680-.3.113278988=.38.540692mV; no double hysteresis.  
C95: 10n COG +/- .5%, 30ppm/C over 100C; allocate + / -10nA external CT leakage.  
tcharge,min=10n*.95*.997*1.17/(1.35uA=10)-1.842676ms > 2.4ms RES minimum.  
tcharge,max=10n*1.05*1.003*1.29/(90u=10n)=15.4627458ms; NOT total-delay max.  
R75 MR pull-up: R=9801.10201ohm; allocate + / -50uA total adverse MR current.  
MR high margin=3.3*1.51819680-50uA/10201=.435459504V; sink <- .396190439mA.  
Conditional DC/startup design, not fast-collapse or SDRAM-quiescence proof.  
Full derivation, assumptions, sourcing and Python: design/reset_coordination_tps3890.md.
```







VLOW=(VCC/RP/Ileak)/(1/RPU+1/(RS+Rcontact)).
Worst stressed VLOW<0.342V; correlated Vih margin>0.315V.
VHIGH>=3-2.2uA*10201=2.9775578V; Vih margin>0.57755V.
Icontact>=3/(10201+1020.6)-2.2uA >265.14uA.
Open contact=2.97745V: switch minimum load=10uA at 2V.
Peak capacitor discharge<3.6/980.1=3.674mA (<50mA).
Four keys held: 4*(3.6/9801+980.1)+2.2uA<1.345mA.

CMS-009 – FILTER TIMING AND BUTTON BEHAVIOR
C58..C61: TDK C1608X7R1H104K080AA, 100nF, X7R, 50V.
Tolerance / temperature-only capacitance bounds:
 $C_{min}=100n \times 0.90 \times 0.85 = 76.5nF$; $C_{max}=100n \times 1.10 \times 1.15 = 126.5nF$
DC bias and aging excluded; qualify effective capacitance.

Nominal tau_assert=(10k|1k)*100n=0.090909ms.
 $\tau_{\text{assert}} = \tau_{\text{assert}} \cdot \ln[(0.2 - 1/1)/(1 - 1/1)] = 0.193\text{ms}.$
 $\tau_{\text{release}} = -10k * 100n * \ln[(1 - 0.8)/(1 - 1/1)] = 1.54\text{ms}.$
 RC filters noise; it does NOT provide complete debounce.
 Require 20ms continuously stable state; switch bounce <= 10ms.
 IRQ starts wake/debounce, not an immediate page/volume action.

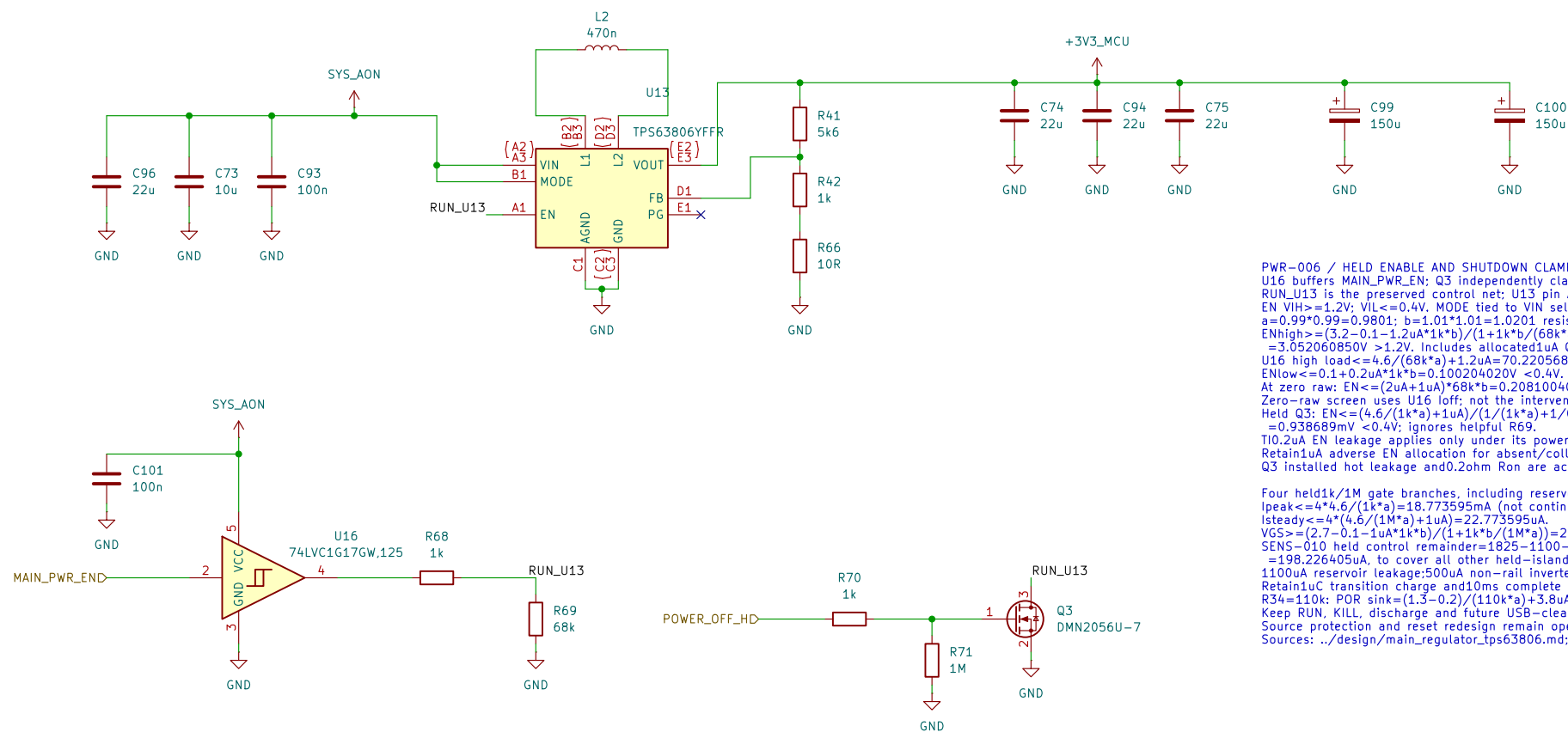
Opposing page volume keys cancel their associated action. Ignore boot-held ordinary actions until release; no auto-repeat from a stuck key. Disable MCU internal pulls.

In hard-off, these four keys cannot start power.

The separate POWER switch controls hardware start/forced-off. ESD441 is not a 3.3V clamp; residual MCU voltage, series resistor pulse stress and ESD current return need physical tests.

Full corner calculations, Python and wake-register contract:
 ../design/camera_storage_interfaces.md CMS-009.

MAIN DIGITAL SUPPLY: TPS63806 / PWR-006
3.272277 V nominal; forced PWM; held shutdown clamp.
Divider and bypass calculations: design/main_regulator_tps63806.md
Reset coordination (#846), source protection and qualification remain open.
DEVELOPMENT SCHEMATIC – NOT FOR POWER-UP OR FABRICATION.



```

PWR-006 / HLD ENABLE AND SHUTDOWN CLAMP
U16 buffers MAIN_PWD_EN; Q3 independently clamps EN during raw collapse.
RUN_U13 is the preserved control net; U13 pin A1 is now EN.
EN VIH>=1.2V; VIL<=0.4V. MODE tied to VIN selects forced PWM.
En=0.99*0.99=0.9801; b=1.01*1.01=1.0201 (resistor factors)
VH=(3.2-0.1-1.2uA*1k*b)/(1-0.9801)=0.96V
VIL=(3.05-0.060850V)/0.99. Included alloted U13 off leakage.
U16 high load=<4.6/(68k*1)+1.2uA=70.220568uA <100uA.
ENlow=0.1+0.2uA*1k*b=0.100204020V <0.4V.
At zero raw: EN<=(2uA+1uA)*68k*b=0.208100400V.
Zero-raw screen uses U16 loff; not the intervening undefined region.
Held Q3: EN<=(4.6/(1k*1)+1uA)/(1/(1k*1)+1/0.2ohm)
=0.9386mV <0.4V; ignores held U16.
T10.2uA EN leakage applies only under its powered test conditions.
Retain U16 adverse EN allocation for absent/collapsing raw.
Q3 installed hot leakage and 0.2ohm Rn are acceptance allocations.

```

```

Four held1k/1M gate branches, including reserved USB-clear:
Ipeak<=4*(1k^2/(1k^2a))=18.773595mA (not continuous ratings).
Isteady<=4*(4*(1M^2/(1M^2a))+1UA)=22.773595mA.
VG55=(2.7-0.1*(1mA^2/(1k^2b)))+(1+1k^2b/(1M^2a))=2.506278V.
SENS=high-level control, remainder=862.5mA/22.773595
=198.226405UA, to cover all other held-island loads.
1100UA reservoir leakage:500UA non-rail inverter-input allocation.
RetainLTC transition charge and10ms complete response allocation.
R34=110k: POR sink=(1.3-0.2)*(110k^2a)+3.8uA=14.003041UA.
KILL=KILL: POR sink=charge and discharge=1000uA, separate.
Source protection and reset redesign remain open; no power-up approval.
Source.../design/main_regulator_tps63806.m; SENS=010 revision.

```

PWR-Vom / FEEDBACK AND LOCAL PASSIVES
 $V_{nom} = 5.00 \times (1 + 5600 / (1000 + 10)) = 3.27227228 \text{ V}$
R41: 0.02%, 5ppm/C; R42: 0.01%, 2ppm/C; R66: 1%, 200ppm/C.
Installed boards: initial "TCR(100C)"aging"assembly, then ohms.
 $R41 = 5600 \times (1 + (-0.0002)) \times (1 + (-0.0005)) \times (1 + (-0.001))$
 $\times (1 + (-0.0005)) \div (-0.02) = 5587.6629237, 5612.349243 \text{ ohm}$
 $R42 = 1000 \times (1 + (-0.0001)) \times (1 + (-0.0002)) \times (1 + (-0.001))$
 $\times (1 + (-0.0005)) \div (-0.01) = 998.190970, 1001.810970 \text{ ohm}$
 $R66 = 10 \times (1 + (-0.01)) \times (1 + (-0.02)) \times (1 + (-0.01)) \div (-0.05)$
 $= 9.954980, 10.455020 \text{ ohm}$ one combined aging/assembly 1%.
 $V_{in} = 5.00 \times (1 + 14160 / (R41 \times R42 \times R66)) = 5.00 \times (1 + 14160 / 1 \times 1 \times 1)$
 $V_{in} = 5.005 \times (1 + 14160 / (R42 \times R66 \times 1)) = 5.005 \times (1 + 14160 / (1 \times 1 \times 1))$
 $V_{static} = 3.226819680, 3.318012496 \text{ V (FPWM only)}$
One combined + / -75mV regulation/ripple/transient/routing allowance:
 $V_{complete} = 3.151819680, 3.393012496 \text{ V}$: qualification required.
SDRAM static VOH margin = $2.4 - 0.73 = 3.93012496 \text{ V} = 24.891253 \text{ mV}$.
This is not measured S/I/noise headroom or reset qualification.

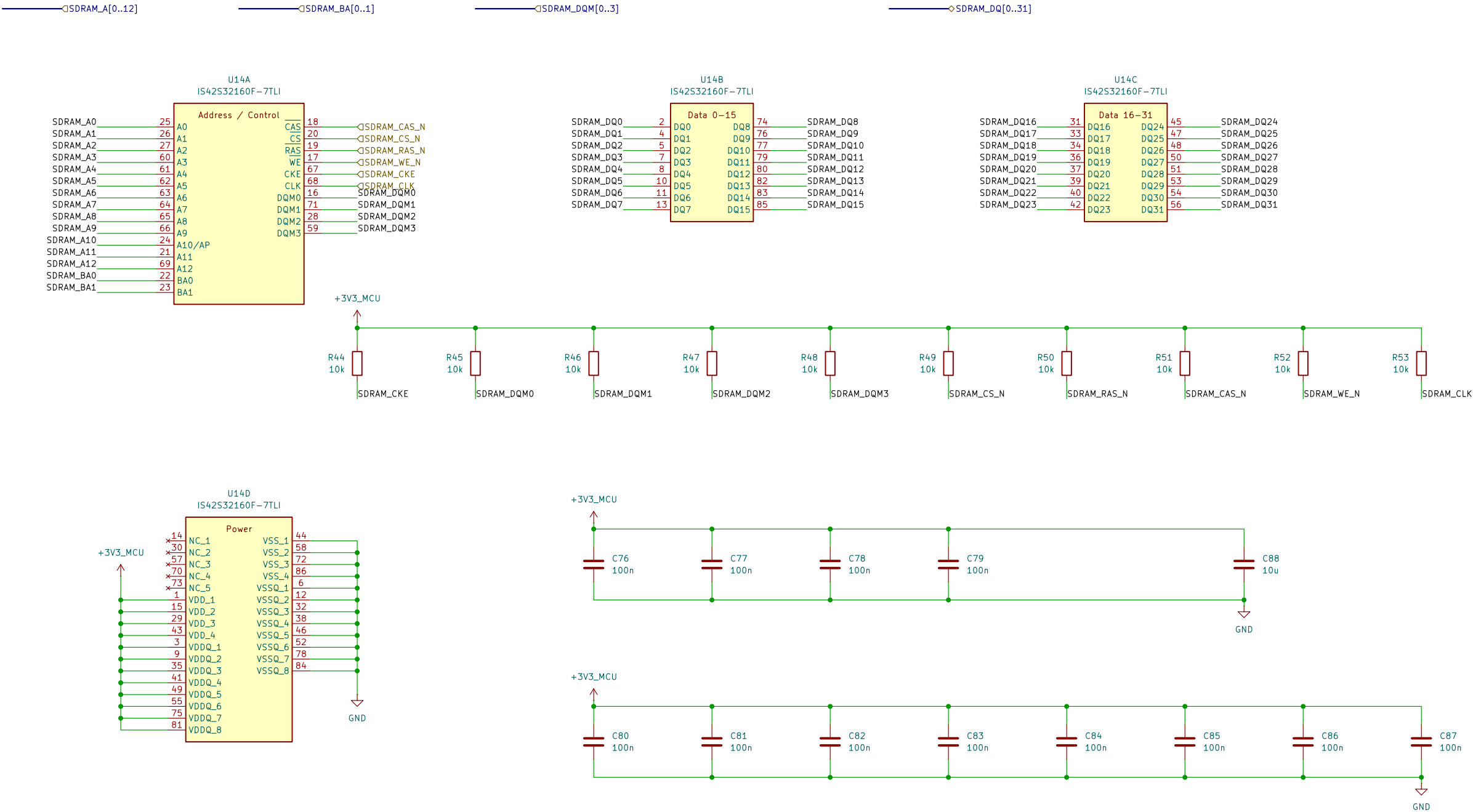
12: 470nH +/-20%; catalog spread alone is insufficient.
Screen incoming0.412..0.518uH; allocated operating+/-10%
= +/-3708..0.5698uH versus T137..0.57uH. Quality hot/current.
C74/C75/C94: *322uF*0.8*0.85*0.6=26.928uF > T121uF
C96: 22uF*0.8*0.85*0.6=8.976uF > T14uF input minimum.
The capacitor combined resistance is not guaranteed.
Samsung typical output bias retention at3.393012496V-90.221554%.
Input bias, aging, mounted impedance and startup need verification.
Full assumptions, sources and executable Python: click this note.

PWR-006 / LDA, STORAGE AND SHUTDOWN
Allocation: 1.80A operating; 2.25A cold start screen, not measured load.
Pwr: 3.393012496e+1, 8=6.10742439372A.
Thermal: 2.544759372A, 2=2.544759372A.
Losses: 0.678602499W, 2=0.678602499W.
TJ90C efficiency: loss=0.678602499W.
TJ90C=70C+78.8C/0.678602499W=123.474C.
2.544759372A is the final PCB thermal resistance.
TJ90C=70C+78.8C/3.393012496e+1=120.179037499W.
Converter limit, source limit, thermal and thermal envelope need validation.

PWR=006 storage checkpoint (nominal, not effective minimum):
Direct main469.21uF + C43 through FB1 10uF + key filters0.4uF
=479.61uF cold; radio-on add±0.2uF =>489.81uF.
C99/C100 remain±150uF each. C94 adds22uF to the output.
Old bootstrap/private VCC/VyF capacitors were not main-rail storage.
Whole-main ceiling remains1mF; reconcile later peripheral additions.
Cout lower acceptance and distributed PDN/startup are separate checks.

```
SENS-010 discharge revision: Rmax1+0.42110hm: return/injection<=+2.5mA.
Includes1.5mA key return +1mA other backfeed: Vinf=0.028552750V.
t=1mF+11.42110hm*ln((3.6-Vinf)/(0.09-Vinf))+10ms+10ms
=66.3987575ms total, below475.794309ms screen.
Includes1.0ms delay+10ms ramp/discharge+10ms delay+10ms below90mV.
Held budget:1.825mA control +0.817mA reverse;IUC transition reserve.
198.226405uA control remainder must cover other held-island loads.
SYS_AON sensor protection, loaded startup and fast faults remain open.
RSN_002 / RADIO-019: release margin remains conditional on rail/path bounds.
No motor power or sensor current correlation or correlation action are complete.
Full assumptions: ../design/main_regulator_tps63806.mt
shutdown revision/Python: ../design/sensor_alternative.md SENS-010.
```

64 MiB SDRAM | ISSI IS42S32160F-7TLI
Signal and reset—default circuits | CMS-010.
Startup, SI/timing and hardware qualification remain open.



CMS-010 | SDRAM power and bypass
VDD and VDDQ share +3V3_MCU (3.0 to 3.6 V).
C76-C79: 4 x 100 nF for the four VDD pins.
C80-C87: 8 x 100 nF for the eight VDDQ pins.
C88: 10 uF local bulk; Cnom = 12 x 0.1 + 10 = 11.2 uF.
Included in the PWR-003 1 mF whole-rail limit.
Effective C, placement and PDN impedance require qualification.

CMS-010 | R44-R53: 10k reset defaults
Rail 3.0..3.6V: +/-1%, 100ppm/C. |dT| <=100C.
Rmin=10k*0.99*0.99=9801 ohm.
Rmax=10k*1.01*1.01=10201 ohm.
Ioff=5uA SDRAM+1uA MCU+1uA board=7uA.
VHIGHmin=3.0-7uA*10201=2.928593V (>2.0V VIH).
Board leakage and temperature screen require qualification.

CMS-010 | Pull load and power states
Sink leakage: 5uA SDRAM+1uA board=6uA.
Isink=3.6/9801+6uA=0.373309mA (<0.374mA).
Pmax=3.6^2/9801=1.322314mW per pull.
All ten LOW: 10*3.6/9801=3.673095mA rail:
10*3.6^2/9801=13.223140mW resistor heat.
Reserve 4mA within the 250mA SDRAM allocation.
CKE/DQM and CS_N default HIGH; CLK HIGH is not a clock.
Switched-rail pulls; self-refresh actively holds CKE LOW.
Startup, dynamic I/O, SI and shutdown qualification OPEN.

Sheet: /64 MiB SDRAM/
File: sdram.kicad_sch

Title: RA8P1 E-reader - 64 MiB SDRAM

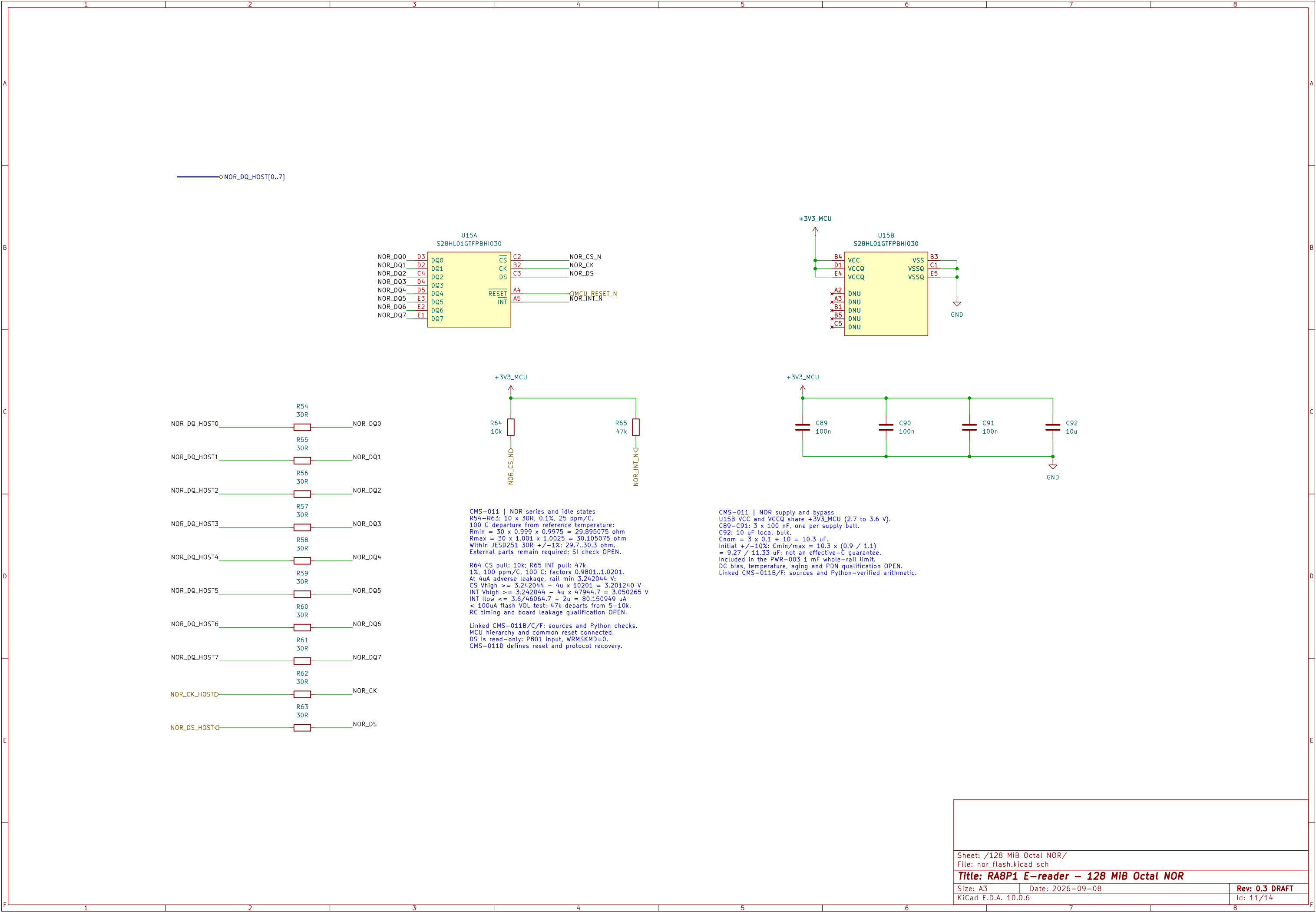
Size: A3

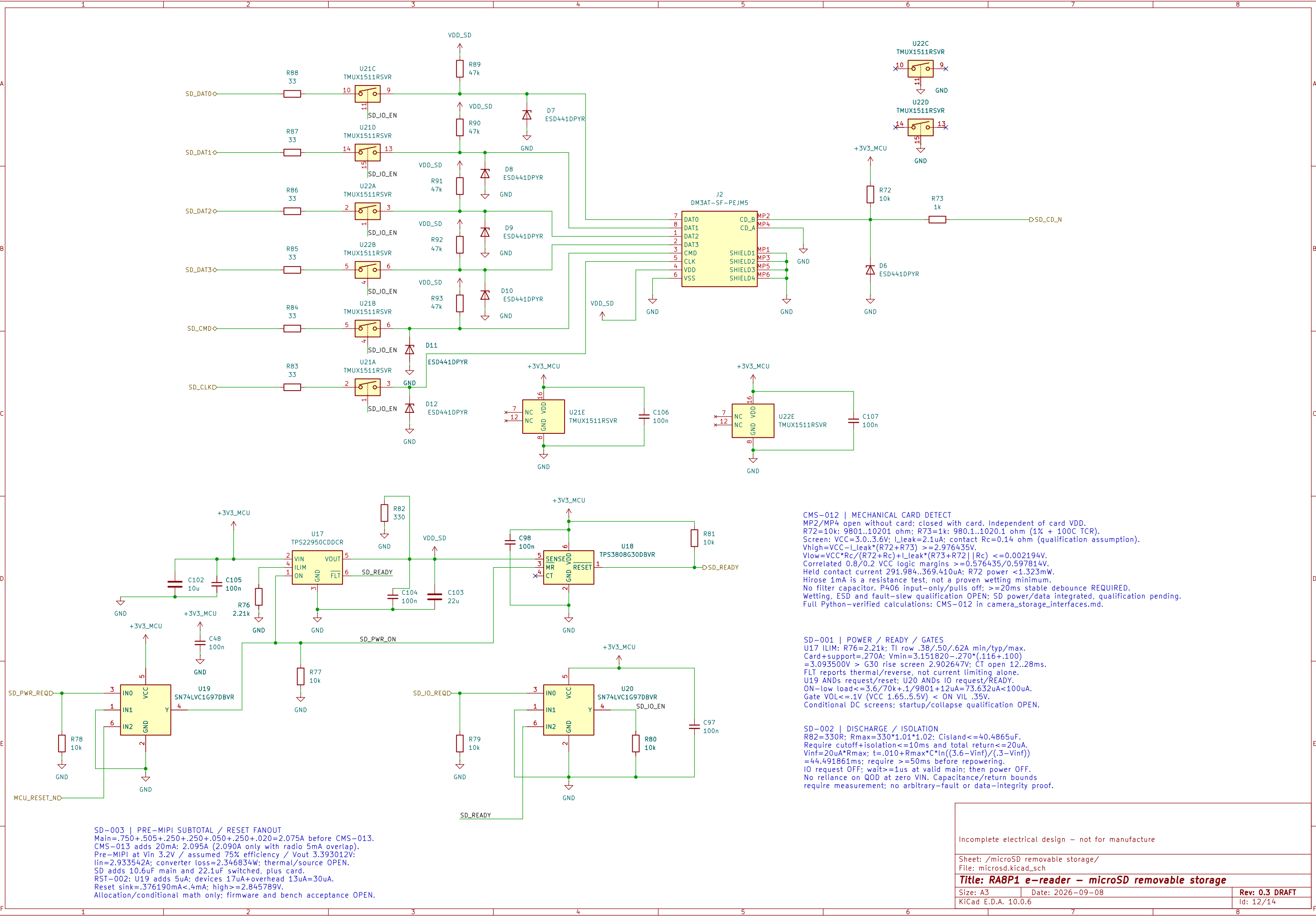
Date: 2026-09-08

Rev: 0.3 DRAFT

KiCad E.D.A. 10.0.6

Id: 10/14





CMS-012 | MECHANICAL CARD DETECT
MP2/MP4 open without card; closed with card. Independent of card VDD.
R72=10k; 9801..10201 ohm; R73=1k; 9801..1020.1 ohm (1% + 100C TCR).
Screen: VCC=3.0..3.6V; I_{Leak}=2.1uA; contact R_c=0.14 ohm (qualification assumption).
V_{high}=VCC-I_{Leak}*(R72+R73) >=2.976435V.
V_{low}=VCC*R_c/(R72+R_c)+I_{Leak}*(R73+R72)|R_c <=0.002194V.
Correlated 0.8/0.2 VCC logic margins >=0.576435/0.597814V.
Held contact current 291.984..369.410uA; R72 power <1.323mW.
Hirosi 1mA is a resistance test, not a proven wetting minimum.
No filter capacitor. P406 input-only/pulls off; >=20ms stable debounce REQUIRED.
Wetting, ESD and fault-slew qualification OPEN; SD power/data integrated, qualification pending.
Full Python-verified calculations: CMS-012 in camera_storage_interfaces.md.

SD-001 | POWER / READY / GATES
U17 ILIM: R76=2.21k; TI row .38/.50/.62A min/typ/max.
Card+support=.270A; V_{min}=3.151820-.270*(.116+.100)
=3.093500V > G30 rise screen 2.902647V; CT open 12..28ms.
FLT reports thermal/reverse, not current limiting alone.
U19 ANDs request/reset; U20 ANDs IO request/READY.
ON-low load<=3.6/70k+.1/9801+12uA=73.632uA<100uA.
Gate VOL<=.1V (VCC 1.65..5.5V) < 0N VIL .35V.
Conditional DC screens; startup/collapse qualification OPEN.

SD-002 | DISCHARGE / ISOLATION
R82=330R; R_{max}=330*1.01*1.02; Cisland<=40.4865uF.
Require cutoff+isolation<=10ms and total return<=20uA.
V_{inf}=20uA*R_{max}; t=.010+R_{max}*C*ln((3.6-V_{inf})/(.3-V_{inf}))
=44.491861ms; require >=50ms before repowering.
IO request OFF; wait>=1us at valid main; then power OFF.
No reliance on QOD at zero VIN. Capacitance/return bounds
require measurement; no arbitrary-fault or data-integrity proof.

SD-003 | PRE-MIPI SUBTOTAL / RESET FANOUT
Main=.750+.505+.250+.250+.050+.250+.020=2.075A before CMS-013.
CMS-013 adds 20mA: 2.095A (2.090A only with radio 5mA overlap).
Pre-MIPI at Vin 3.2V / assumed 75% efficiency / Vout 3.393012V:
I_{in}=2.933542A; converter loss=2.346834W; thermal/source OPEN.
SD adds 10.6uF main and 22.1uF switched, plus card.
RST-002: U19 adds 5uA; devices 17uA+overhead 13uA=30uA.
Reset sink=.376190mA<.4mA; high>=2.845789V.
Allocation/conditional math only; firmware and bench acceptance OPEN.

Incomplete electrical design – not for manufacture

Sheet: /microSD removable storage/
File: microsd.kicad_sch

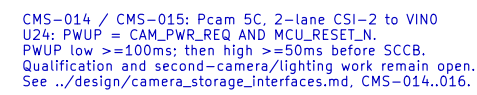
Title: RA8P1 e-reader – microSD removable storage

Size: A3 Date: 2026-09-08

Rev: 0.3 DRAFT

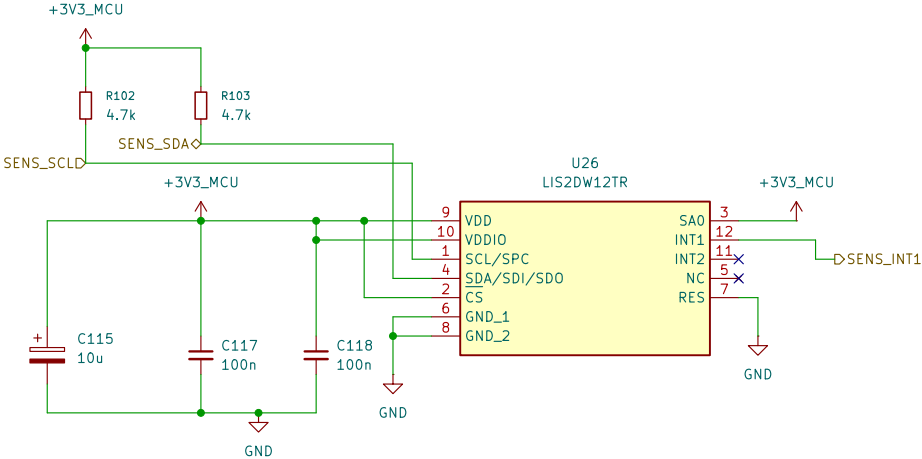
KiCad E.D.A. 10.0.6

Id: 12/14



Sheet: /Pcam 5C camera/ File: camera.kicad_sch		
Title:		
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.6		Id: 13/14

ORIENTATION SENSOR – WIP
U26: LIS2DW12TR; ST DS11811 Rev 9 pin map.
VDD pin 9 / VDDIO pin 10: +3V3_MCU; CS / SA0 high for I2C address 0x19.
C117/C118: local 100n bypass; GND pins 6/8 and RES pin 7 grounded.
C115: 10u/16V aluminum EEE–FM1C100R; place near VDD pin 9 with C117.
R102/R103: 4.7k SCL/SDA pull–ups; obsolete ADXL VREG network removed.
Host connected: SCL P410/K13; SDA P409/R16; INT1 P306/D11.
Wait 20ms after supply valid. Power–cycle below 100mV for >=10ms.
Pending: sourcing fields, pin modes, timing/power/passive qualification.
See design/sensor_alternative.md; circuit qualification remains open.



Sheet: /Orientation sensor/
File: orientation_sensor.kicad_sch

Title:

Size: A4
KiCad E.D.A. 10.0.6

Date:

Rev:

Id: 14/14